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REVIEW ARTICLE

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Internet of things in disaster management: technologies and uses

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ABSTRACT

In order to aid professionals in disaster management, efficient procedures and technologies are developed and adopted. This paper intends to present the results of a literature review on the use of the Internet of Things (IoT) in disaster management. Even though the IoT approach was defined many years ago, only recently its technologies have been enhanced and acquired greater clarity and scope. Internationally, scientific studies indexed in two databases, Scopus and Web of Science were found, selected and analyzed. Two portfolios of papers were generated: the first for the bibliometric analysis and, the second, leaner, for qualitative analysis retaining the contributions pertinent to the proposed theme. The literature review identified macro-discussions on the IoT in disaster management and applications of more specific devices or technologies. In disaster operations, the IoT is a target theme in the fields of Computer Science, Electronic Engineering and Environmental Engineering, especially useful in disaster mitigation, rescue, response, and recovery. The main positive impact is the facilitated communication between professionals, more robust data and monitoring of data for environmental management, first aid, people tracking, distribution of supplies and medicines, among other purposes.

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Natural disaster; Internet of Things; environmental monitoring; technology transfer

1. Introduction

Extraordinary adverse events have always been a constant part of people's lives from the onset of civilisation. For many years, it was not possible to register those occurrences due to a lack of knowledge and technological support (Drabek & Mcentire, 2003). In literature, one of the first records of a disaster is related to an accident that occurred at Halifax Harbor in Canada, 1917, involving a French ship carrying explosive cargo and a Belgian ship. The accident entailed thousands of deaths and impacts to the community and the environment (Drabek & Mcentire, 2003).

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Adverse events or disasters are caused by climate factors, accidents, or intentional human actions (terrorist attacks or arson, for instance) (Laplante et al., 2016). A few other noteworthy records of disasters include the Bhopal gas accident (in India, 1984), the Chansala mining disaster (in India, 1975), the terrorist attack on September 11th (in the USA, 2001), the Chornobyl nuclear accident (Ukraine, 1986), the tsunami in the Indian Ocean (2004), the earthquake in Nepal (2015), and the Fort McMurray forest fire (in Canada, 2016) (Ray et al., 2017), among others.

Regardless of the cause, overall disasters impact a group of people (victims), the local community, the environment, authorities, and countries in terms of public health, economy, environment, and politics.

Due to the impacts generated by disasters, people endeavour to recover their physical health as much as possible, normalise their day-to-day activities, rebuild houses and other locations, and reestablish agricultural and labour operations. These actions require specialised support from healthcare professionals, public policies, government financial subsidies, and material donations, among other goods and services. In disaster management, sets of actions are applicable to situations of natural disasters or accidents, minimising impacts through emergency assistance; searching for missing victims; supplying food, shelter, and diverse materials; and reconstructing affected locations (Dachyar et al., 2019).

Technology is a key ally in disaster management. In its diverse shapes and aspects, it performs specific roles handled by professionals and technicians. In this context, technology advances entail benefits, such as prevention, care, more efficient treatments, and other advantages through Knowledge and Technology Transfer (KTT). KTT is embedded in the global processes of innovation and product development, from the acquisition of machinery and equipment (Arvanitis et al., 2008), through training and technical support, and to technology enhancements (Dziallas & Blind, 2019; Sosnowski, 2014).

Since the 1990s, investments in KTT centred on digital applications and tools have generated new means of communication between people. A crucial global contribution was the Internet (Li et al., 2012). In turn, a recent approach has emerged: the Internet of Things (IoT), which connects people and objects with more accuracy to the web (Li et al., 2012). The IoT may be defined as the technology of connected devices that exchange data with one another, acting as agile enablers of data, information, and services (Van Der Klauw, 2018).

The purpose of this paper is to present the results of a literature review on the Internet of Things (IoT) applied to a specific area, disaster management. The results were obtained through bibliometrics and the qualitative analysis of relevant scientific papers in the international literature.

The IoT corresponds to a set of information and communication technologies, engineering solutions, and advances in computer sciences. Complex technologies are interconnected and create smart object networks, in which they communicate, exchange information, generate more consistent data, and provide support for strategic decisions and actions. In disaster management area, structured procedures and technologies are essential in prediction, rescue, and post-disaster operations. People, material goods, financial resources, and environmental resources, among others, are affected by natural or man-made disasters that occur around the world. Therefore, exploring technologies that may be useful to enhance disaster management and make operations more efficient is necessary.

Even though it is theoretical, the paper proposes to provide a discussion on an emerging theme, the IoT, which has been recently the focus of analyses. Associated to disaster

management, this study guides the discussion to one of the infinite IoT applications, so as to provide a better comprehension of the subject to professionals and interested parties, besides generating other complementary studies, more in-depth and empirical.

The literature review was conducted in two international databases, namely Scopus and Web of Science, defining a bibliographic portfolio of papers for a bibliometric analysis and, subsequently, qualitative analysis, following formal literature review protocols, according to Pagani et al. (2015, 2017). This paper is structured into four sections. The first contextualises the study and presents the goal. The second and third sections exhibit the methodology and results, respectively. The final section identifies the conclusions.

2. Methodology

This study conducted a literature review in order to fulfil its goal. A review is an integral part of a research project, explicitly revealing the universe of scientific contributions in a given area (Dos Santos, & Candeloro, 2006). The literature review was comprised of the following procedures, as defined by (Pagani et al., 2015, 2017). These procedures were applied in the studies of Bail (2019), Bail et al. (2020), Pagani et al. (2019), Silva et al. (2018), and Silva et al. (2019), among others. The steps for the review were:

Steps 1, 2 and 3 – Establishing the intention of research, preliminary exploratory research, and definition of research syntax: the intention was to explore the combinations 'disaster management' and 'internet of things'. Preliminary exploratory search with keywords in databases Scopus and Web of Science revealed a larger amount of works linked to the themes researched in these databases. The goal of the review was to explore scientific papers indexed in these databases. Keywords combinations were fixed, both inserted in paper titles or abstracts, with no temporal restrictions (all years), and the databases previously mentioned.

Step 4 – Final search in the databases: data collection was done on August 21th, 2020. For this task, the reference manager Mendeley® was used. The bibliometrics analysis with papers was applied before the next steps, so that more data could be included in the study.

Step 5 – Applying filtering procedures: duplicate papers were discarded, employing the reference manager Mendeley®.

Step 6 – Identifying impact factor, year and number of citations: the metrics of the papers (impact factor) were obtained from the Thomson's Reuters/Clarivate Analytics website and Scopus website. The number of citations was obtained from Google Scholar. This information – metrics and number of citations – along with the year of publication, is necessary to calculate the InOrdinatio.

Step 7 – Ranking the papers using the InOrdinatio: in order to rank the papers according to their scientific relevance, the index of ordination – the InOrdinatio Equation (1) – is applied, using an electronic spreadsheet.

$$\text{InOrdinatio} = (IF/1000) + (\alpha \times (10 - (RY - PY))) + (Ci) \quad (1)$$

The elements of the equation are: IF (Impact Factor); α (alpha value – weighting factors ranging from 1 to 10 to be attributed by the researcher, according to the importance

of the novelty, theme); RY (year in which the bibliographic research was developed); PY (year in which the analyzed paper was published); and Ci (number of times the paper has been cited on Google Scholar®) (Pagani et al., 2015); and, Step 8 and 9 – Downloading the selected studies and reading them in full: the bibliometric analysis found the main years of publication, the foremost journals, the areas that encompass the researched theme (the Internet of Things oriented to disaster management), the authors who published about the subject, the authors' countries of origin, and the most frequent terms. Therefore, the bibliometric analysis was done with papers found until Step 4. The qualitative analysis of the papers required more refined filters (Pagani et al., 2015; Silva et al., 2020). Therefore, the generated bibliographic portfolio, previously employed for bibliometrics, was also submitted to the procedures on Steps 5, 6 and 7: reading the titles and abstracts to check the alignment with the research scope and ranking the papers employing Equation (1), and analysing the selected papers by reading them in their entirety and gathering the relevant discussions, respectively.

3. Results and discussion

This Section is divided into two subsections: 3.1 Bibliometrics, which presents quantitative analysis, and; 3.2 Content analysis, which brings qualitative analysis from an in-depth reading. For the quantitative analysis, the tools from the Scopus and Web of Science and Voswiwer® software were applied.

3.1. Bibliometrics

The search in two databases employing the keywords 'disaster management' and 'internet of things' returned 285 documents, of which 151 are journal papers (Table 1).

Figure 1 displays the distribution of papers over the years.

It is noticeable that the discussions about the Internet of Things oriented to disaster management are current; there were 25 papers in 2020 (January-August), 32 in 2019, 17 in 2018, and 10 papers in 2017, especially. The first journal paper on the subject was indexed in the database and published in 2010.

Diverse types of scientific journals published papers, as shown in Table 2. These journals' scopes mainly include Computer Science and Information Technologies, followed by journals that exclusively focus on disaster management.

Based on the number of published papers, the journals that stood out were IEEE Access, with an Impact Factor (IF-2019) of 3.745, Computer Communications 2.816, Sensors (Switzerland) 3.275, and IEEE Internet of Things Journal 9.515. The journals that focused on scenarios and natural and environmental resources were the Natural Hazards 2.419, Sustainability (Switzerland) 2.576, and International Journal of Disaster Risk Reduction 2.896, mainly.

Table 1. Total of papers on the Internet of Things applied to disaster management.

Keywords	Scopus	Web of Science
'Internet of Things' and 'Disaster management'	83 papers	36 papers
Gross total	151 papers	
Total after discarding duplicates	97 papers	

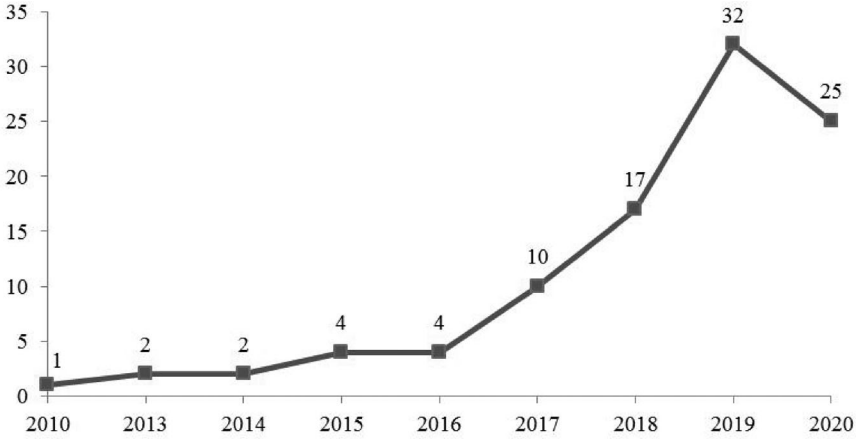


Figure 1. Years and frequencies of published papers on IoT oriented to disaster management operations. Source: Data collected from Scopus and Web of Science databases (2020).

The journals with publications on IoT oriented to disaster management have received papers linked to the various areas of knowledge, such as Computer Science (66 journal papers) and Engineering (49 papers) (engineering multidisciplinary, engineering electrical and electronic, engineering environmental and engineering civil), mainly (Figure 2).

Figure 3 presents the main authors considering the number of publications.

Zaveri, M. A presented four different papers published, followed by the authors Asimakopoulou, E.; and Bessis, N., with three papers each. The authors are based in India, China, and the United States, as displayed in Figure 4.

The terms that best define the papers are displayed in Figure 5, namely Internet, IoT, information technologies, disaster, disaster management, environment, and impact.

3.2. Qualitative analysis

Through reading the titles and abstracts of 97 papers (Table 1) and applying the InOrdinatio (Equation 1), 27 papers were selected for the qualitative analysis (Appendix 1).

Table 2. Main scientific journals.

Journal	Frequency (Number of Papers)
IEEE Access	8
Computer Communications	6
Sensors (Switzerland)	5
BM Journal of Research and Development	3
IEEE Internet of Things Journal	3
Journal of Advanced Research in Dynamical and Control Systems	3
Applied Sciences (Switzerland)	2
International Journal of Distributed Systems and Technologies	2
International Journal Of Disaster Risk Reduction	2
Journal of Natural Disasters	2
Natural Hazards	2
Sustainability (Switzerland)	2
Wireless Communications and Mobile Computing	2

Source: Data collected from Scopus and Web of Science databases (2020).

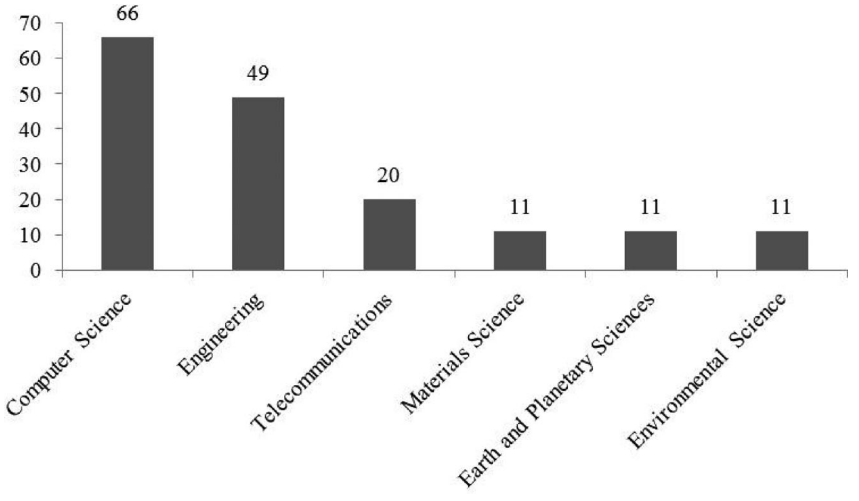


Figure 2. Main areas of published papers on IoT oriented to disaster management. Source: Data collected from Scopus and Web of Science databases (2020).

Papers published in conferences were also adopted as supplementary material. The purpose of the next subsections is to provide knowledge about Internet of Things (IoT) technologies, and later, relying solely on other researchers, we explain how IoT fits in the area of disaster management.

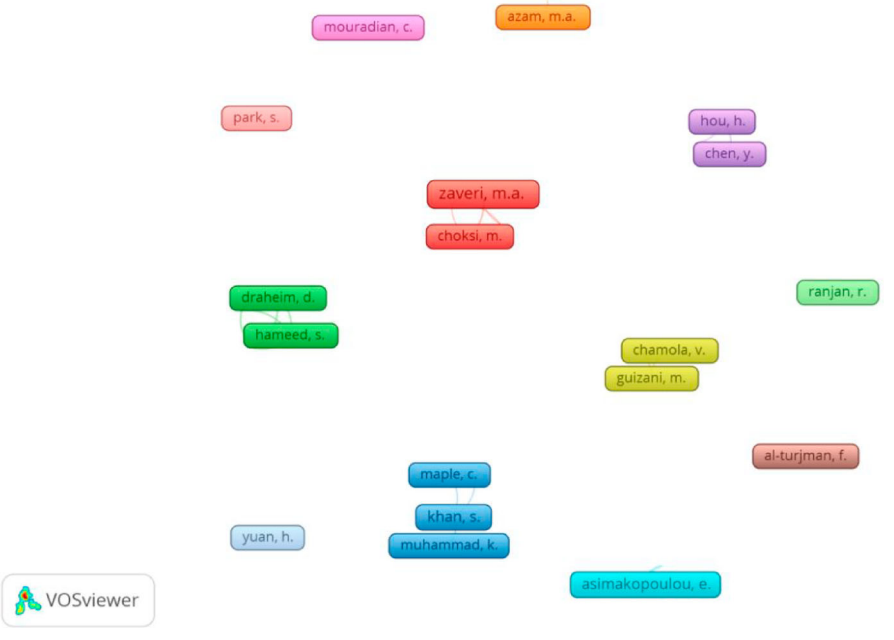


Figure 3. Main authors with papers on on IoT oriented to disaster management. Source: Data collected from Scopus and Web of Science databases (2020).

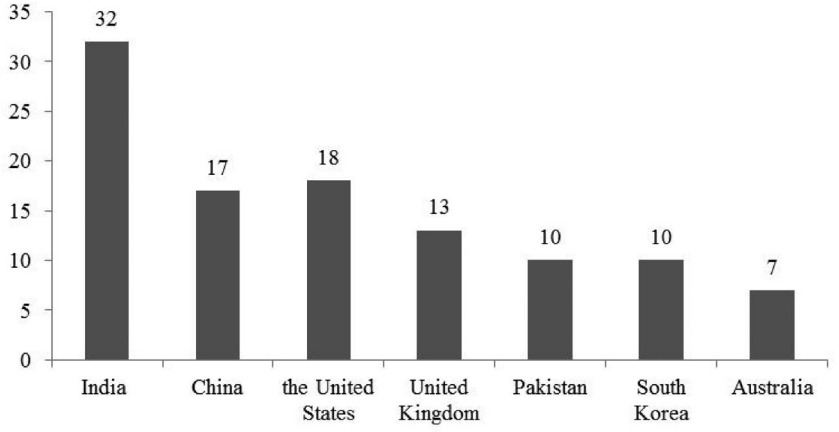


Figure 4. Countries with participation in publications on IoT oriented to disaster management. Source: Data collected from Scopus and Web of Science databases (2020).

3.2.1. Internet of things concepts and structure

The term Internet of Things (IoT) was coined by the researcher Kevin Ashton in 1998, revolutionising the concepts of objects of day-to-day use since then. The objects were connected to the wide world web, transforming the methods of connectivity and communication between objects, machines, products, and people (Sinha et al., 2019). Perera et al. (2013) reported the evolution of the Internet until the arrival of the IoT, such as network (communication between two computers), the Internet (virtual network communication between computers), mobile Internet (expansion of

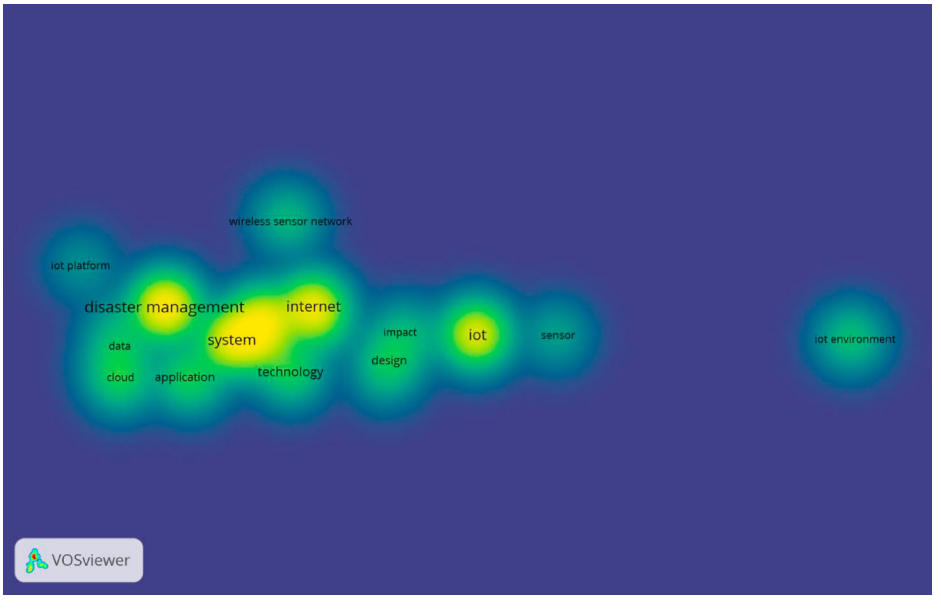


Figure 5. Main terms inserted on papers. Source: Data collected from Scopus and Web of Science databases (2020).

communication through devices and computers simultaneously), and Internet of Things (interconnected objects; for example, computers, mobiles, sensors, machines, and houses as well as peoples).

The evolution of the Internet began with the connection between two computers. Then, connections between computers and the web arose, followed by mobile devices connected to the web, people connected to computers, mobile devices through social networks, and the IoT connecting day-to-day objects to the Internet (Perera et al., 2013). According to Zanella et al. (2014), the IoT aims to render the Internet even more immersive and widespread, seeing that it allows greater interaction between an array of devices, such as appliances, security cameras, monitoring sensors, actuators, displays, and vehicles, among others.

In other words, the IoT is a network of physical objects (things) that contain embedded technologies to communicate or interact with their internal states or with the external environment (Hung, 2017). It is a new technological paradigm that behaves as a global network of machines and devices capable of interacting with each other (Lee & Lee, 2015). According to Da Xu et al. (2014), IoT is regarded as a connection of countless devices or as an infrastructure of technologies. Several devices are inserted into this connection and may be seen as indispensable to allow this interaction, such as Radio-Frequency Identification (RFID), which automatically identifies, tracks, and monitors objects through microchips installed in products and wireless networks.

In summary, five technologies are widely employed to implement IoT-based products: Radio-Frequency Identification (which allows automatic identification and data capture using radio waves, a tag, or a digital reader), wireless sensor networks (which consist of devices equipped with autonomous sensors spatially distributed to monitor physical or environmental conditions), middleware (software layer or a set of interjacent sub-layers between the technological and application levels), cloud computing (model for on-demand access to a shared set of configurable resources), and IoT application software (which allows device-device and human-device interactions reliably and solidly) (Lee & Lee, 2015). The IoT structure basically consists of sensors, actuators, firmware, sensor network, Middleware and its frameworks, and other technologies (Perera et al., 2013).

The IoT may be associated with three pillars, which are related to the capability of smart objects to (i) be identifiable, (ii) communicate, and (iii) interact, generating networks of objects interconnected among themselves or to users (Miorandi et al., 2012); (Al-Fuqaha et al., 2015) presented and exemplified six central elements that are essential to provide IoT functionality. They exemplified the IoT elements in:

(1) Identification: Objects configured within the network may use public IPs and, seeing that they are not globally exclusive, require an identity. A few of the identification methods employed for the IoT are electronic product codes (EPC) and ubiquitous codes (uCode), which present IPv4 or IPv6 addresses.
(2) Sensing: the data collection employs sensors and devices embedded into physical objects, such as smart sensors, wearable sensing devices, embedded sensors, actuators, and RFID tags, among others.
(3) Communication: Physical objects are connected in order to provide intelligent services. With that purpose, they employ communication protocols such as Radio-

Frequency Identification (RFID), Near Field Communication (NFC), Ultra-Wide Bandwidth (UWB), Bluetooth, Bluetooth Low Energy (BLE), IEEE 802.15.4, Z-wave, WiFi, WiFiDirect, and Long-Term Evolution Advanced (LTE-A).

(4) Computation: One of the IoT interfaces is composed of applications and their hardware platforms, such as Arduino, FriendlyARM, Intel Galileo, Raspberry PI, and Gadgeteer, among others. In contrast, software platforms are employed to provide IoT functionalities via operational systems, including TinyOs, LiteOs, RiotOS, and Android.
(5) Service: They may be categorised into identity-related (shipping), information aggregation (smart grid); Collaborative-Aware (smart home), and Ubiquitous (smart city). Each presents specific application purposes.
(6) Semantics: it allows knowledge to be generated through the collection and extraction of data from physical objects such as machines for decision-making processes.

The IoT has gone from the connection between cell phones or computers to objects with complex functions, such as digital self-service kiosks, self-driving cars and trains (passenger services and maintenance), and drones (for food delivery, medicine delivery, and security) (Hung, 2017).

The connection of goods, processes, and people enables the capture of data and events, allowing preventive measures to be managed (Hung, 2017). According to Yang et al. (2010), by allowing new means of communication between people and things and between things themselves, the IoT renders data and information management even more efficient. The IoT is designed to establish device management, communication, and monitoring, conducting heterogeneity, interoperability, distributed processing, and real-time data analyses (Ray & Turuk, 2017).

3.2.2. IoT applications in disaster management

Disaster management encompasses a set of structured processes that incorporate the operations of planning, management, and action application at any stage of the disaster, that is, mitigation, rescue, response, and recovery. The central goal is the integration of interrelated processes that may offer more efficient methods to analyze, monitor, predict, and manage disasters (Velev et al., 2018). In order to ensure the exchange between the institutions, experts, and other members involved in disaster management, information and communication technologies that have roused collective interest include Big Data, Cloud Computing, social media tools, mobile applications, sets of intelligent sensors, and structures of the Internet of Things (Velev et al., 2018).

Regarding the Internet of Things geared towards disaster management, studies in the literature propose models that mainly focus on more specific IoT approaches or technologies. Table 3 describes the central approaches and procedures of Internet of Things applications in disaster management.

Studies on the Internet of Things are geared towards models, analyses, and discussions of scenarios and cases of the management of disasters. Table 4 describes the main IoT technologies and components applicable to disaster management.

As observed in the studies from international literature, the IoT may not prevent the occurrence of disasters, but it can be useful in their management via the systems of prediction and notices, rescue operations, environment monitoring, and victim aid (Laplante et al., 2018 and Amiri et al., 2019). The disaster management becomes more efficient as

impact data are obtained from different sources, from strategic and reliable locations, which make data closer the reality. Therefore, the IoT is applicable in these cases, seeing as it maximises the communication between objects, with simultaneous exchanges of data from real environments (Adeel et al., 2019).

In disaster management, a crucial contribution of IoT technologies is providing real-time data support to the experts who need to make decisions before, during, or in response to disaster recovery (Arslan et al., 2017). According to these authors, IoT technologies also help to broaden disaster-related knowledge in terms of its characteristics and the impacts generated. Experts may benefit with technological assistance for their work, benefiting also the population.

The interconnection of devices and other objects to the network, composed of sensors, mobile devices, machines, computers, and satellites, among others, has the potential to advance the actions of risk identification and assessment, adoption of preventive measures, mapping of affected areas and locations, and other actions that constantly require data and information due to the need for resource allocation and strategies (Sumanth & Ome, 2018).

IoT provides support for data collection, big data analysis, data monitoring, information sharing, and mutual collaboration between people / professionals involved in disaster management (Lee et al., 2018). IoT has applicability in different contexts of disaster management, as described in Table 5.

When a specific region is equipped with sensors, devices, microcontrollers, and software applications – all connected to the network with remote access to their results – it becomes a self-protecting and self-monitoring environment (Sumanth & Ome, 2018).

Considering the advantages, the European Commission, through IoT research, aims to expand new technologies, promoting the expansion of satellite communication (involving the activation of emergency sound devices), real-time communication, images and accurate location of the disaster, creation of a management system for missing people, among countless other possible practices (Yang et al., 2010).

Technology has limitations, but it also has the potential to help in the planning, management, and analysis of disaster management operations, as well as reducing their effects. The IoT approach incorporates effective data collection strategies and the ability to share those data. It can manage complex decision support systems, conducting necessary services in a more accurate, organised, and smart manner (Sinha et al., 2019).

4. Conclusions

IoT is a theme that, although it has been in circulation for many years, only recently has acquired greater scope and clarity in its approaches and architectures. The area of greatest interest in research is Computer Science. It represents a high weight, both in numbers of authors and journals linked to it, succeeding the Engineering. Within disaster management, this area is predominant, in addition to areas Environmental Engineering and Environmental Science.

As observed in the literature review, IoT technologies aid professionals from several areas, including healthcare, materials logistics for those affected by disasters, environment monitoring, and global data control, among others. These technologies are highly important for those scenarios, seeing that the objects, distributed through

Table 3. Main studies based on the contributions presented.

Author	Title	Study Focus	Procedure
Laplante et al. (2016)	Standards for the Internet of Things: A Case Study in Disaster Response	Describes an Organization for the Advancement of Structured Information Standards (OASIS) of connections interlinked to the IoT as the central tool for preventing natural disasters.	IoT enhancements in cities connected to patients in hospitals.
Ray et al. (2017)	Internet of Things for Disaster Management: State-of-the-Art and Prospects	Systematic research on disasters and the IoT, following the main protocols that aim to neutralise the disaster setting, infrastructure in uncovered layers, data management, communication, and security.	Description of numerous infrastructures that aid in disaster management, coupled with: wireless energy sensors; micro-controllers of information security; Transport Layer Security (TLS); the Nano Internet Protocol (NanoIP), and several other devices that seek rapid connections between organs of security and emergency.
Dachyar et al. (2019)	Inventory management design for a rapid disaster relief, towards internet of things (IoT) potential	The authors point out three levels of actions (pre, during, and post-disaster). The IoT presents the capabilities of: support in visualising the disaster scenario; minimising risks through monitoring possible locations and risks in advance; information systems interconnected in real-time and timely assistance; finding missing people, dynamic situations, and other information in real-time.	Implementation of business process reengineering (BPR) integrated into the IoT in logistics in case of disasters.
Sinha et al. (2019)	Impact of internet of things (IoT) in disaster management: a task-technology fit perspective	Use of the IoT in accessing victim data by doctors and emergency professionals. Focus on natural disasters.	TTF (Task-Technology Fit) approach to analyze the importance of adopting the IoT in disaster management.
Shah et al. (2019)	Towards Disaster Resilient Smart Cities: Can Internet of Things and Big Data Analytics be the Game Changers?	Developing systems integrated in the IoT considering natural disasters. This work involves the four research actors (government, university, industry, and community).	IoT model in integrated systems connections, capturing information from Smart City and disaster management (fires in buildings, pollution levels, evacuation routes).
Al-Turjman (2019)	Cognitive routing protocol for disaster-inspired Internet of Things	Proposal of data structure in large-scale networks, suggesting tools that offer greater availability of access, less consumption, agility and trust, modifications and improvements in the capture of data transmissions in the case of disasters.	Identification of mechanisms that reduce the amount of energy consumed by the programmes interconnected with the IoT and failure management in systems.
Han et al. (2019)	Harnessing the power of crowdsourcing and Internet of Things in disaster response	Application of the IoT in several segments. 'Crowdsourcing and the Internet of Things' – employing social means of communication and Radio-Frequency Identification (RFID) technology to aid disaster management.	Identification of existing tools and apps on how to act in moments of disasters.
Amiri et al. (2019)		Comparing two data routing tools and their functions: rescue operations, damage scale, information gathering,	-

(*Continued*)

Table 3. Continued.

Author	Title	Study Focus	Procedure
	DABPR: a large-scale internet of things-based data aggregation back pressure routing for disaster management	disaster management, smart routing, smart data transmission.	
Qin et al. (2018)	Research on the technological architectural design of geological hazard monitoring and rescue-after-disaster system based on cloud computing and Internet of things	Study conducted in China in mountainous regions that have over 210 thousand occurrences a year. Monitoring, alerts, pre/post-disaster interventions.	IoT with state-of-the-art information, cloud computing, integration of RFID system, decision-making model. It promotes, through technology, reduction in disasters with decision-making before catastrophes.
Lee et al. (2018)	An Internet of Things System Architecture for Aiding Firefighters in the Scene of Disaster	Presenting an Internet of Things system architecture that may be configurable and applicable to firefighting and rescues in several disaster situations.	-
Laplante et al. (2018)	Could the Internet of Things Be Used to Enhance Student Nurses' Experiences in a Disaster Simulation?	The study's central idea originated on the September 11th terrorist actions. It demonstrates the importance of preparing student nurses for acting during catastrophes, disasters, and terrorist attacks.	Health system linked to the IoT based on the configurations (in short and long-term) to aid the community, such as apps to track victims, monitoring traffic during the disaster, and training.
Gaire et al. (2018)	Internet of Things (IoT) and Cloud Computing Enabled Disaster Management	Providing an app for supporting the life cycle of disaster management data (dissemination of alerts, data management).	Sensors implemented in several risk areas to collect environmental data. It seeks to centralise and canalise the data emitted by the tools, filtering them by relevance and priority.
Hu et al. (2014)	Coal Mine Disaster Warning Internet of Things Intrusion Detection System Based on Back Propagation Neural Network Improved by Genetic Algorithms	Study focused on the strong and weak points of the Back Propagation Neural Network (BPNN) tool (anti-intrusion method/virtual safety), employed to recognise intruders in the IoT.	Detecting errors and enhancing a virtual data collection system.
Zhou (2014)	Research of geological disaster intelligence monitoring system based on internet of things	Presents an intelligent monitoring method for geological disasters based on the Internet of Things.	Proposal of a method which uses IoT technology and carries out the real-time online monitoring of various parameters of landslides and debris flow in the environment.
Yuliandoko et al. (2018)	Design of flood warning system based IoT and water characteristics	Presents an IoT technology project to monitor and control a system for the early detection of flooding disasters.	Microcontroller was used and connected with sensors for the analysis of water level, speed, and flow. These sensors have been configured and used in river dam experiments, simulating probable cases of flooding.
Cui (2020)	Deployment and integration of smart sensors with IoT devices detecting fire disasters in huge forest environment	Reports a case study where intelligent sensors were deployed in a forest environment to monitor and record environmental impacts.	Each IoT device connected to the network communicates with others via 4G Internet and web interfaces. In the operation, IoT devices detect atmospheric variables, gases, and pollutants in the environment. The collected data is analyzed and processed using algorithms – Convolution Neural Network.

Table 4. Disaster management and the IoT technologies employed.

Disaster type	Disaster Management Category Approached	Subcategory	IoT Components and Technologies	Source
General disasters	Prevention, planning, assessment, recovery.	Early warning, notification, real-time data analysis, remote monitoring of the affected area, victim location, and knowledge management about disasters for experts.	Wireless sensor networks (WSNs), internet control protocol, and Open Systems Interconnection models (OSI).	Ray et al. (2017)
	Action planning and assessment.	Detection and analysis of the disaster magnitude.	Radio-Frequency Identification sensors (RFID), infrared sensors, ground sensors, Global Positioning Systems (GPS), Wireless Sensor Networks (WSNs), Device-free Passive Localization (DfPL).	Deak et al. (2013)
	Recovery	Providing health services to the victims. Real-time monitoring of data related to their vital signs (blood pressure, heart rate, blood oxygen level, and breathing rate). Data sharing.	Set of IoT-enabled medical sensors, mobile device, ESP8266 data module, Linux server.	Misbahuddin et al. (2018)
Natural disasters in general	Recovery.	Immediate and suitable aid to the victims. Rescue operations.	Medical equipment with wireless communication, sensors, gateway system, Cloud Computing.	Sinha et al. (2019)
Geological disasters	Prevention of greater damage.	Online, real-time monitoring of several parameters of landslides and debris flow. Alerts.	IoT technologies, General Packet Radio Service (GPRS) communication technologies, µC/OS-II embedded operating system.	Zhou (2014)
Floods	Prevention.	Precise alarm system.	Ultrasonic and infrared sensors, ESP-8266 microcontroller, mobile device, specific software, internet services.	Yuliandoko et al. (2018)
	Prevention and planning.	Access, exchange, and use of data from sensors from distinct providers (acting organisations in disaster management).	GPS-enabled tracking sensors, weather surveillance radars, lightning detection sensors, rain gauges, urban area cameras.	Alamdar et al. (2017)
Earthquakes	Recovery.	Post-earthquake victim rescue.	Emergency wireless communication networks, such as Zigbee and Wideband Code Division Multiple Access (WCDMA).	Zhang et al. (2014)
Earthquakes and floods	Prevention.	Alarm systems.	Sensors, wireless internet networks, and mobile devices.	Pawar and Katiravan (2019)
Environmental fires	Recovery.	Post-disaster victim rescue, control of environmental fires.	Sensors and Wireless Sensor Networks (WSNs). Open Microsoft Technology Center (MTC) platform. Fog computing.	Kavitha et al. (2019)
Building fires	Assessment and recovery.	Rapid rescue of building dwellers. Information support to the emergency team.		Park et al. (2018)

(*Continued*)

Table 4. Continued.

Disaster type	Disaster Management Category Approached	Subcategory	IoT Components and Technologies	Source
			Sensors and actuators, gateway, internet server, database, big data analytics concepts, augmented reality technologies, mobile devices, and specific applications.	
Volcanic eruptions	Prevention and planning.	Alerting the population.	Sensors to collect the parameters of temperature, humidity, CO_2 gas, wind speed, wind direction; rain monitoring; wireless internet; application of data control panel.	Damayanti et al. (2020)
Accidents in factories	Prevention and planning.	Disaster early-warning and prediction system.	Geographic information about the coal mine surface, the underground geologic information, and the mine's production system are monitored in real-time through sensors. Employing big data technology, the data volumes collected are processed.	Chong-mao et al. (2015)
Radiation	Assessment and recovery.	Disaster monitoring and coordination of emergency responses.	Intelligent sensors, Wireless Sensor Networks (WSNs), actuators, internet server, and mobile devices.	Lin and Liaw (2015)
General disasters in closed organisations and environments	Action planning in the event of disasters.	Ordered evacuation of people and shelter availability.	Mobile device, wireless communication infrastructure, internet server, Cloud computing platform.	Xu et al. (2018)

Source: Research data (2020).

Table 5. Main applications of IoT oriented to disaster management.

Application context	Description
Knowledge about the disaster	The ability to detect, in a distributed manner, natural phenomena such as temperature, wind, and precipitation, integrating heterogeneous data (Miorandi et al., 2012). More consistent decision making in disaster anticipation or action planning (Miorandi et al., 2012). Minimization and prevention of disaster risks (monitoring potential disasters via satellite communication, early alert systems, and the use of social media to raise awareness) (Sinha et al., 2019).
Providing direct assistance to victims	It presents solutions in the monitoring of patient parameters such as body temperature, blood pressure, and breathing activity, among others (Miorandi et al., 2012). Through advanced services, it is possible to optimise a city's physical infrastructure (vehicles, hospitals, homes and shelters) and society's quality of life (Miorandi et al., 2012). IoT-based in the disaster management will require effective and reliable interoperability from all the systems, including the personal traceable devices of victims, such as phones. Other devices in reach will also be able to interact with each other, including vehicles, companies, and smart buildings (Laplante et al., 2018). Emergency response (real-time communication for timely measures and assistance), and making easier recover in cases of disasters (online search for missing people and funds management systems), among other situations. (Sinha et al., 2019).
Supplies management, such as medicines, food and water and clothing	Applicability in stock management, product distribution, deliveries, and tracking (Miorandi et al., 2012). Creating smart transportation systems, tracking vehicles, and road and traffic (Velev et al., 2018) among others.
Environmental management	Interconnected objects provide the support for a better understanding of the environment, decision-making processes, and the application of actions in critical scenarios of disaster occurrences (Zelenkauskaite et al., 2012).

Source: Research data (2020)

diverse settings, communicate and generate data and information in real-time. The components and technologies identified in this paper are described in Chart 1.

We can conclude that the Internet used through wireless sensors and mobile devices are the main topics when it comes to disasters and IoT. These technologies are of great help for disaster prevention, recovery planning, and damage assessment. Nevertheless, it is matter of a conjoint work of universities, public governance, and aid organisations.

IoT is still an incipient subject, complex and containing rather specific approaches, as illustrated by the complex architectures of sensors, software, protocols, and smart devices, among other specifics.

The main contribution of this paper is to provide knowledge about IoT, both in the macro context and in a more specific area, disaster management. One of the motivating reasons for this research is that disasters occur all over the world, and they affect people, government – political and economic aspects, and environments and demand supplies and efforts.

Despite the contributions, this research has also some limitations. The first one is the number of bibliographic databases we used in the research. The second one is that our contribution is based only on journal papers.

Based on these limitations, we suggest that further researches could explore additional bibliographic databases, and also different types of contributions, for instance, conference papers. Also, we suggest that practical studies on the application of IoT technologies

Description
• Wireless sensor networks (WSNs), internet control protocol, and Open Systems Interconnection models (OSI).
• Radio-Frequency Identification sensors (RFID), infrared sensors, ground sensors, Global Positioning Systems (GPS), Wireless Sensor Networks (WSNs), Device-free Passive Localization (DfPL).
• Set of IoT-enabled medical sensors, mobile device, ESP8266 data module, Linux server.
• Medical equipment with wireless communication, sensors, gateway system, Cloud Computing.
• IoT technologies, General Packet Radio Service (GPRS) communication technologies, μC/OS-II embedded operating system.
• Ultrasonic and infrared sensors, ESP-8266 microcontroller, mobile device, specific software, internet services.
• GPS-enabled tracking sensors, weather surveillance radars, lightning detection sensors, rain gauges, urban area cameras.
• Emergency wireless communication networks, such as Zigbee and Wideband Code Division Multiple Access (WCDMA).
• Sensors, wireless internet networks, and mobile devices.
• Sensors and Wireless Sensor Networks (WSNs). Open Microsoft Technology Center (MTC) platform. Fog computing.
• Sensors and actuators, gateway, internet server, database, big data analytics concepts, augmented reality technologies, mobile devices, and specific applications.
• Sensors to collect the parameters of temperature, humidity, CO_2 gas, wind speed, wind direction; rain monitoring; wireless internet; application of data control panel.
• Geographic information about the coal mine surface, the underground geologic information, and the mine's production system are monitored in real-time through sensors. Employing big data technology, the data volumes collected are processed.
• Intelligent sensors, Wireless Sensor Networks (WSNs), actuators, internet server, and mobile devices.
• Mobile device, wireless communication infrastructure, internet server, Cloud computing platform.

Chart 1. IoT Components and technologies.

in disaster management can be searched, considering that discussions regarding the IoT oriented to disasters and correlated themes are still incipient.

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Appendix 1. Portfolio of articles selected for qualitative analysis.

N	Author	Title	Year	Number of Citations	Impact factor (2018)	InOrdinatio values of Methodi Ordinatio
1	Al-Turjman, F.	*Cognitive routing protocol for disaster-inspired Internet of Things*	2019	61	5,768	161,01
2	Ray, P., Mukherjee, M. and Shu, L.	*Internet of Things for Disaster Management: State-of-the-Art and Prospects*	2017	75	4,098	155,00
3	Sinha, A., Kumar, P., Rana, N., Islam, R. and Dwivedi, Y.	*Impact of internet of things (IoT) in disaster management: a task-technology fit perspective*	2019	30	2,284	130,00
4	Han, S., Huang, H., Luo, Z. and Foropon, C.	*Harnessing the power of crowdsourcing and Internet of Things in disaster response*	2019	9	2,284	109,00
5	Shah, S., Seker, D., Rathore, M., Hameed, S., Ben Yahia, S. and Draheim, D.	*Towards Disaster Resilient Smart Cities: Can Internet of Things and Big Data Analytics Be the Game Changers?*	2019	4	4,098	104,00
7	Choksi, M., Zaveri, M.A.	*Multiobjective Based Resource Allocation and Scheduling for Postdisaster Management Using IoT*	2019	3	1,396	103,00
7	Visu, P., Lakshmanan, L., Murugananthan, V. and Cruz, M.	*Software-defined forensic framework for malware disaster management in Internet of Thing devices for extreme surveillance*	2019	2	2,766	102,00
8	Sathish Kumar, J., Zaveri, M.A.	*Resource Scheduling for Postdisaster Management in IoT Environment*	2019	1	1,396	101,00
9	Amiri, I., Prakash, J., Balasaraswathi, M., Sivasankaran, V., Sundararajan, T., Hindia, M., Tilwari, V., Dimyati, K. and Henry, O.	*DABPR: a large-scale internet of things-based data aggregation back pressure routing for disaster management*	2019	0	2,405	100,00
10	Dachyar, M., Yadrifil and Fahreza, I.	*Inventory management design for a rapid disaster relief, towards internet of things (IOT) potential*	2019	0	0	100,00
11	Uma Priyadarsini, P. and Sriramya, P.	*Disaster management using evidence-based interactive trust management system for wireless sensor networks by Internet of Things*	2019	0	2,189	100,00
12	Yang, J., Hou, H., Chen, Y. and Han, L.	*An Internet of Things based material delivery model for disaster management in libraries*	2019	0	1,256	100,00
13	Xu, X., Zhang, L., Sotiriadis, S., Asimakopoulou, E., Li, M. and Bessis, N.	*CLOTHO: A Large-Scale Internet of Things-Based Crowd Evacuation Planning System for Disaster Management*	2018	8	9,515	98,01
14	Qin, L., Feng, S. and Zhu, H.	*Research on the technological architectural design of*	2018	6	0	96,00

(*Continued*)

Continued.

N	Author	Title	Year	Number of Citations	Impact factor (2018)	InOrdinatio values of Methodi Ordinatio
		geological hazard monitoring and rescue-after-disaster system based on cloud computing and Internet of things				
15	Lee, H., Hong, S. and Lee, K.	*An Internet of Things system architecture for aiding firefighters in the scene of disaster*	2018	5	0	95,00
16	Laplante, N., Laplante, P. and Voas, J.	*Could the internet of things be used to enhance student nurses' experiences in a disaster simulation?*	2018	3	0	93,00
17	Raj Gaire et al.	*Internet of Things (IoT) and Cloud Computing Enabled Disaster Management*	2018	3	0	93,00
18	Deak, G., Curran, K., Condell, J., Asimakopoulou, E. and Bessis, N.	*IoTs (Internet of Things) and DfPL (Device-free Passive Localisation) in a disaster management scenario*	2013	48	0	88,00
19	Laplante, P., Voas, J. and Laplante, N.	*Standards for the Internet of Things: A Case Study in Disaster Response*	2016	11	2,766	81,00
20	Choi, C.-H., Hong, Y. and Lee, J.	*A study on IoT (Internet of things)-based disaster detection and prevention system*	2016	1	0	71,00
21	Li, Z., Wang, T., Gong, Z. and Li, N.	*Forewarning technology and application for monitoring low temperature disaster in solar greenhouses based on Internet of Things*	2013	19	0	59,00
22	Hu, Y., Sun, L., Yu, S., Huang, J., Wang, X. and Guo, H.	*Coal mine disaster warning internet of things intrusion detection system based on back propagation neural network improved by genetic algorithms*	2014	3	0	53,00
23	Meng, X. and Feng, L.	*Research on application of internet of things technology to earthquake prevention and disaster reduction*	2014	0	0	50,00
24	Shi, Z., Han, H., Yin, S. and Xu, Q.	*Tentative plan of application of the mobile internet of things in mountain disaster emergency communication system*	2014	0	0	50,00
25	Zeng, C.	*Mountain torrent disaster monitoring and early-warning system based on internet of things*	2014	0	0	50,00
26	Zhou, A.-H.	*Research of geological disaster intelligence monitoring system based on internet of things*	2014	0	2,295	50,00
27	Ju, H.	*Research on the framework of high performance the internet of things based on multilevel disaster recovery*	2012	0	0	30,00